

Effect of rewiring for a sandpile model on a directed network

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ABSTRACT

Several studies have considered sandpile dynamics not over regular grids, but over networks. In this case, avalanches redistribute grains not between neighboring sites in a geometrical sense, but between connected sites, in a topological sense. However, depending on how nodes are connected, grains may never leave the system, preventing energy release. In this work, we study the simplest case, the BTW model in one and two dimensions, rewiring the nodes so that at every rewiring step, the energy release is always possible, and study avalanche statistics as a function of rewiring. In the 1D case, a transition is observed in the Gini coefficient of the load distribution per node at about 85% the number of possible rewirings, a transition which is not evident with other measures, such as the size distribution of avalanches or the mean distance between nodes in the network. In the 2D case, energy release follows a power law even when the grid is fully rewired, while the Gini coefficient, unlike the 1D case, decreases at a steady rate, with a smoother transition. The effect of network size N is studied, finding that there is a transition for the Gini coefficient at the thermodynamic limit $N \rightarrow \infty$ for both the 1D and 2D cases, transition which is also observed in the betweenness centrality, but not in other topological measures. Finally, the dependence of the results with the load per rewiring iteration, and the avalanche threshold is studied.

1. Introduction

Self-organized criticality (SOC) provides a fundamental framework for understanding the emergence of complexity in dynamical systems. Sandpile models [1] are the paradigmatic representation of SOC behavior. Traditionally, these models consider a regular grid of cells where an external driver slowly accumulates energy. When the load on a given cell reaches a specific threshold, it is redistributed to neighboring cells, triggering a cascade of relaxations (avalanches) until all loads are below the threshold.

The ubiquity of SOC is reflected in its applications across vastly different natural and social phenomena. Beyond the well-known energy release processes in magnetized space plasmas such as the Earth's magnetosphere [2] and solar flares [3–6] SOC signatures are found in numerous macroscopic systems. For instance, in geomorphology, the flow transported through river networks exhibits directed SOC dynamics via sediment cascades [7,8]. Similarly, the accumulation and stochastic release of magma in volcanic eruptions follow a directed flow of self-organized criticality [9]. Other examples include fractal electrical discharges, where intracloud dynamics release energy following power-law distributions [10], and economic and financial systems, where fluctuations in buying and selling occasionally trigger critical events such as market crashes [11].

While traditional sandpiles operate on Euclidean lattices, the real-world systems such as those mentioned above, are often characterized by heterogeneous, complex structures. Consequently, the generalized case of SOC dynamics over complex networks has gained significant attention [12,13]. Numerical and theoretical studies have extensively explored sandpile models on specific

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static topologies, such as random [14,15], scale-free [12,16], and Apollonian networks [17], as well as broader theoretical developments [18]. It is widely accepted that network topology fundamentally modifies avalanche statistics and energy dissipation mechanisms.

A particularly challenging scenario arises when the underlying topology is not static but has its own dynamics. Prime examples of such phenomena are found in the aforementioned magnetized plasmas, where avalanches are closely tied to magnetic reconnection processes that continuously modify the topology of the magnetic field [19–22]. Recently, we investigated the Lu-Hamilton cellular automaton model for solar flares [23], demonstrating how progressive network rewiring introduces a macroscopic transition in avalanche statistics, leading to a nonlocal development of dissipation events.

In addition to dynamical modeling, complex network analyses for magnetized plasmas have proven invaluable for studying the emergent topology resulting from observational data, such as solar flares [24,25], or sunspots and active regions [26,27]. It has even been proposed that these observational network methods can be applied to forecast the solar cycle [28] and to reveal the underlying 22-year Hale cycle [29]. Whereas these data-driven approaches involve building networks from observed events, there remains the issue of how transport properties evolve as the topology of the underlying network changes.

Motivated by this, the present work aims to study the structural transition of a simple directed sandpile model as it evolves from a regular grid into a complex network. A challenge when selecting an arbitrary underlying network to model a SOC system, where energy is loaded and dissipated, is the potential emergence of isolated nodes (which indefinitely accumulate grains) or closed topological loops (which trap energy and prevent global dissipation). To overcome this, we apply a controlled rewiring process to the sandpile model, one link at a time, ensuring that energy release pathways are always mathematically guaranteed. We systematically investigate the resulting features of the stationary state of the sandpile, studying the distribution of load across the network by means of the Gini coefficient, and characterizing the network by means of structural metrics such as degree and betweenness centrality. Ultimately, we apply size scaling analysis to demonstrate how these topological alterations trigger a phase transition in the macroscopic dissipation of the system.

The paper is organized as follows: in Section 2, the directed sandpile model and the network construction process are described. The numerical and topological results are presented in Section 3, followed by an analysis of size effects in Section 4. Finally, our findings are summarized and discussed in Section 5. In addition, robustness to threshold modifications and driving load is presented in the Supplementary Material.

2. Model

We consider the sandpile model in one and two dimensions, as explained below.

2.1. 1D model

We start from the basic proposal by Bak et al. [1], where grains are loaded on a grid, and avalanches occur when the height difference between neighboring sites reaches a certain threshold U . We consider a simple one-dimensional grid of N sites, at each timestep $Q = 1$ grains are loaded on the grid, on a site which is selected randomly.

Since the problem is analogous to a system subject to a constant gravitational potential, one can define the energy of the i th site, with n_i grains, as

$$E_i = \frac{n_i(n_i - 1)}{2} , \quad (1)$$

the condition to trigger an avalanche as

$$E_i - E_{i+1} \geq U , \quad (2)$$

the total energy of the system as

$$E = \sum_{i=1}^N E_i , \quad (3)$$

and the dissipated energy in an avalanche as the difference between the energy in the sandpile after E_{after} and before the avalanche E_{before} ,

$$\Delta E = E_{\text{after}} - E_{\text{before}} .$$

To study the sandpile dynamics over a network, we first represent the system described above with a directed complex network, as seen in Fig. 1. Here, each node represents a site in the grid, and directed edges represent the direction of the avalanche once the threshold condition is met. Thus, the node i is connected to the node $i + 1$, so that, if the threshold condition is satisfied at node i , then a grain is moved from node i to node $i + 1$. The last node, which has no outgoing edges, is the node where energy is finally released from the system during the avalanche. A grain loaded into the system, once the avalanche occurs, eventually leaves the system at this node.



Fig. 1. Network representation of the BTW model, where each node i delivers energy to node $i + 1$, and node $n = 6$ releases energy from the system.

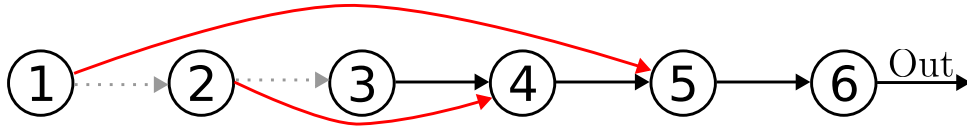


Fig. 2. Rewired network configuration and its corresponding M_2 matrix, after two rewirings.

This network can be represented by an adjacency matrix full of zeros, except for ones above the diagonal. For instance, if the network has $N = 6$ nodes, the adjacency matrix would be

$$M_0 = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

Given this configuration, at each iteration a random node i is selected, then one grain is added to it, and the threshold condition (2) is tested. If it is satisfied, one grain is moved in the direction of the outgoing edge at i , that is, from node i to node $i + 1$, and the avalanche proceeds until Eq. (2) is false at every site.

Running a sandpile on this configuration leads of course the same results as the BTW model, as shown in Section 3. However, we intend to consider the case where the topology of the network changes, so that grain displacement does not necessarily occur on consecutive nodes, or equivalently, where adjacent nodes are defined by the network connections. To investigate how topological transformations alter the system's configuration and avalanche statistics, we systematically modify the network by rewiring one link at a time until all connections have been updated. This process is executed sequentially: we first rewire the outgoing connections of the first node, then proceed to the second, and so forth. For each rewiring operation, the new target node is chosen uniformly at random, subject to a strict constraint: its index must be greater than that of the original target node.

This topological restriction is critical, as arbitrary rewiring would destroy the fundamental properties required for the SOC dynamics. For instance, in the linear chain depicted in Fig. 1, if the directed edge from node 5 to node 6 were replaced by an edge pointing backwards to node 4, then node 6 would be isolated, meaning that grains dropped elsewhere in the network could never reach it. Furthermore, such backward connections introduce closed topological cycles (loops). Consequently, an avalanche reaching node 4 could circulate indefinitely within this loop, trapping the energy and preventing the system from properly dissipating it through the boundaries.

In our treatment, we select the nodes starting from the first node in the chain, down to the second to last node in the change, and rewire it to a randomly selected node, in such a way that the energy can always be released from the grid, and there are no accumulation points. This is done by following two rules: (a) an edge is modified only once, and (b) if the old edge goes from node i to $i + 1$, the new edge goes from node i to a node $j > i + 1$. This is enough to ensure that the grid is never broken.

As an example of a rewired grid, see Fig. 2, where two rewirings have been performed, with the resulting adjacency matrix

$$M_2 = \begin{pmatrix} 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}. \quad (4)$$

Regardless of where grains are added to the sandpile, energy is always released at node 6.

In general, valid rewirings, which do not separate the system into isolated subnetworks, and always have a single node where energy leaves the system, can be easily represented in terms of the adjacency matrix as follows:

$$\begin{aligned} M_{i,i+1} &= 0, \\ M_{i,\text{rnd}[i+1,N]} &= 1, \end{aligned} \quad (5)$$

where node i has been selected for rewiring, and $\text{rnd}[i + 1, N]$ represents a random integer j such that $i + 1 < j \leq N$. The initial linear configuration of N nodes, can be rewired at most $N - 2$ times.

We can see the pseudocode in Alg. 1,

Algorithm 1 Directed Rewiring Algorithm (Preserving Direct Acyclic Graph (DAG) Structure)**Require:** Initial Network $G = (V, E)$ (Directed Grid), Target Rewiring Count n_{target} **Ensure:** Rewired Network G' (Directed Acyclic Graph, constant total degree)

```

1:  $count \leftarrow 0$ 
2: while  $count < n_{target}$  do
3:   Select a source node  $i \in V$  uniformly at random
4:   if  $i$  is a sink node (boundary) then
5:     continue
6:   end if
7:   Identify the current neighbor  $k$  such that  $(i, k) \in E$ 
8:   Select a candidate target node  $j > i \in V$  uniformly at random
9:   if  $i \neq j$  and  $j \neq k$  then
10:    if There is no directed path from  $j$  to  $i$  then
11:       $E \leftarrow E \setminus \{(i, k)\}$ 
12:       $E \leftarrow E \cup \{(i, j)\}$ 
13:       $count \leftarrow count + 1$ 
14:    end if
15:  end if
16: end while
17: return  $G(V, E)$ 

```

▷ Constraint 1: Avoid self-loops and existing connections
 ▷ Constraint 2: Ensure Acyclicity (DAG)
 ▷ Remove old link
 ▷ Add new link

The last node does not have an outgoing edge, and every grain reaching it is discarded unconditionally, whereas the second to last node has only one connection (to the last node), so both nodes cannot be rewired.

Notice that this algorithm generates all possible directed acyclic graphs with N nodes, such that $N - 1$ nodes have out-degree equal to 1. (The exception is the last node.) More general acyclic graphs could be built by adding new connections instead of only rewiring, or by more general algorithms [30], but here we focus on incremental departures from the original BTW model. Also, by construction, at every step in the algorithm the resulting network has a tree structure, two arbitrary nodes being connected by a unique path. This tree-like structure arises naturally, as all paths must reach the last node where energy is released, and all connections must be directed from a certain n th node to an m -node, with $m > n$.

The resulting network structure is also interesting, as various systems which exhibit SOC behavior, also involve direct transport on a fractal-like structure with a shape similar to directed acyclic graph, as energy flows to a sink. Such is the case, for instance, of river basins, magma flows, and electrical discharges [10,31–33]. The concept of directed acyclic graphs has also gained traction in economics and finance, where they are used to model causal structures in price discovery and risk propagation [34–36].

In principle, the rewiring algorithm described above could be used for any type of network, preserving the number of connections and nodes. However, in our case, outgoing nodes are not changed from the initial configuration, then a random choice of nodes to rewire can break the network, affecting energy release. Further improvements could be made by accepting rewirings only if they do not break the network, or adding new outgoing modes, but we will consider the simple model proposed to study it systematically.

Now, given a certain rewired configuration, we iterate the sandpile model, adding grains at random nodes at each iteration, and study the features of the resulting stationary state as the underlying grid departs from the simple linear chain, as a function of the number of rewirings.

2.2. 2D model

We start from the unperturbed adjacency matrix is

$$M_0 = \begin{pmatrix} 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \quad (6)$$

which is equivalent to the 2D grid of the BTW model, as shown in Fig. 3:

In this case, energy leaves the system at the edges. Following the strategy for the 1D model, we now rewire nodes in such a way that energy always flows downhill, and that energy can be released from the system at the same nodes as in the regular 2D grid. This means that connections leaving the network are not modified.

Thus, the same two rewiring rules as in the 1D case, Eq. (5), can be used.

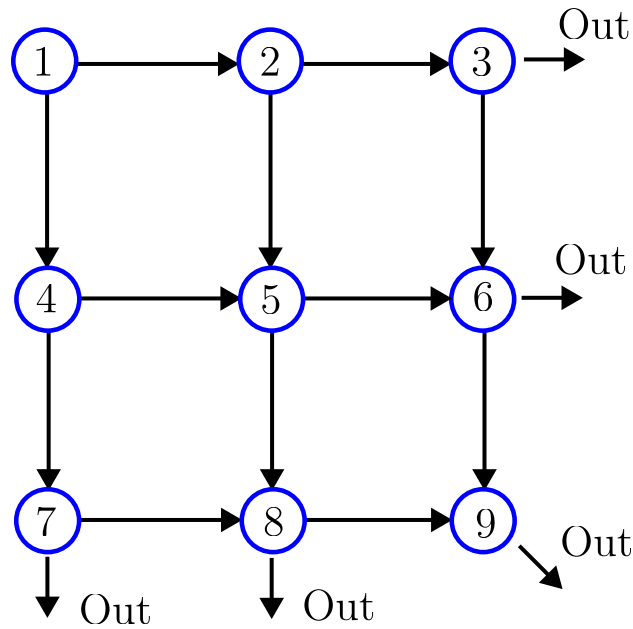


Fig. 3. Graphical representation of the initial adjacency matrix (6).

For instance, after two rewirings the system can be given by an adjacency matrix such as the following:

$$M_2 = \begin{pmatrix} 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}. \quad (7)$$

This can be represented by Fig. 4.

3. Results

3.1. 1D case

In the following, we study a system with $N = 256$ nodes, an energy threshold $U = 2$ to trigger an avalanche, and where $Q = 1$ grains are loaded on the sandpile at a time. For every run, we consider 10^8 iterations, which is more than enough to reach the stationary state. And we follow the system for 10^5 iterations once the transient has ended.

For the initial linear chain, the stationary state is reached after about 133 000 iterations, as shown in Fig. 5, where the total energy E in the system is plotted as a function of iteration time t .

The shape of the stationary state is, as expected, a simple configuration where the i th node has $U = 2$ grains more than the $i + 1$ -th node (Fig. 6).

The Probability Distribution Function (PDF) of energy release events at the stationary state is essentially constant (Fig. 7) which is consistent with the fact that during an avalanche a grain simply moves through the pile until the last site, and avalanches of all sizes can occur with equal probability, depending only on the site where the grain is loaded, which is itself chosen from a uniform distribution. Thus, avalanche size statistics are trivial in this case [1].

Since the distribution of grains is not uniform across nodes in the stationary state, it makes sense to calculate the “inequality” of its distribution, through the Gini coefficient, which turns out to be $G \simeq 0.33$ for the linear chain. This will give us an easy way to characterize the shape of the stationary state as rewirings are introduced.

We now rewire the sandpile, as described in Section 2, and study the effect on the stationary state. The first observation is that, as rewirings are made, the average energy in the system decreases, and the number of iterations needed to reach the stationary state decreases. This is shown in Fig. 8, where curves of the energy in the sandpile after each iteration are shown for six different values of the number of rewirings n_R .

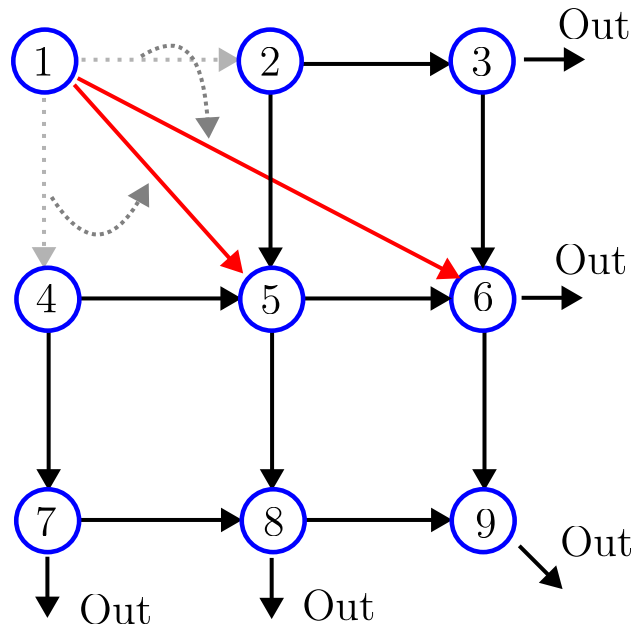


Fig. 4. Rewired network configuration according to the M_2 matrix in Eq. (7), after two rewirings.

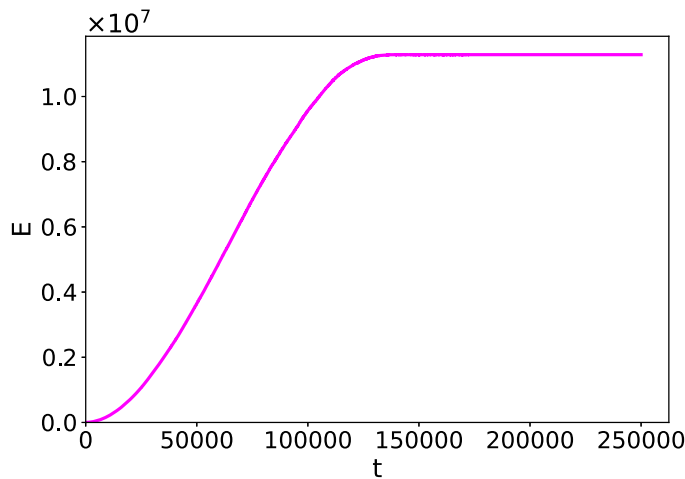


Fig. 5. Energy of the system for the non-rewired 1D case (linear chain) as a function of iteration time t , showing the transition from the transient to the stationary state.

Before rewirings, all nodes receive avalanches from the previous one, and they can discharge only to the following node. By rewiring, the paths of discharge are increased, which leads to a shorter time to reach the stationary state. This also decreases the energy of the stationary state: since discharges can follow more than one path, the condition (2) must be met by more pairs of nodes, and since loads to the network are at random sites, the threshold condition can be satisfied earlier than for the linear chain, reducing the average load of nodes.

This can be clearly seen by observing the shape of the stationary states, as shown in Fig. 9, for the same networks represented in Fig. 8.

As expected from the analysis of Fig. 8, rewirings have the effect of reducing the average load of all nodes, since discharges are more likely to occur. It can also be clearly noticed that this occurs first in the first nodes in the chain, given the sequence in which nodes are rewired. Thus, rewirings are enough to destroy the simple configuration of the stationary state BTW model, and which in turn leads to a trivial dynamics, as shown in Ref. [1]. In Ref. [1], a more interesting dynamics is found by considering the bidimensional case. Here, we see an alternative way by rewiring the linear chain, which is another way to provide more discharge paths.

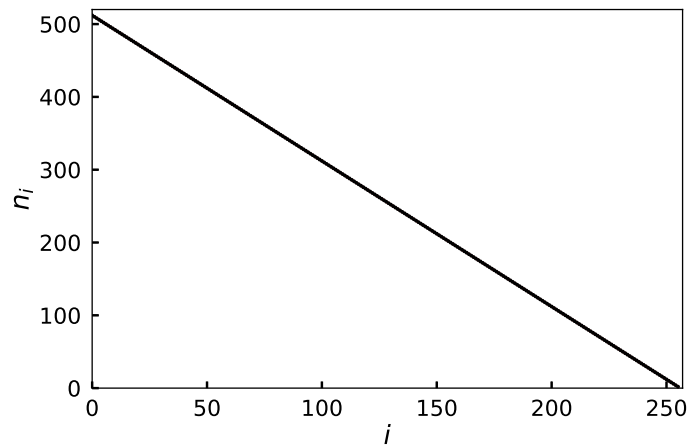


Fig. 6. Number of grains at the i th state at the stationary state for the linear chain, with no rewiring.

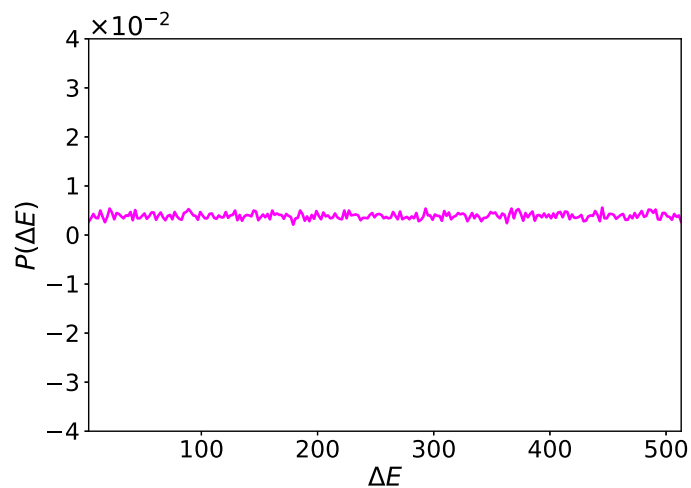


Fig. 7. Probability Distribution Function (PDF) of the energy released ΔE during avalanches for the linear chain, without rewirings.

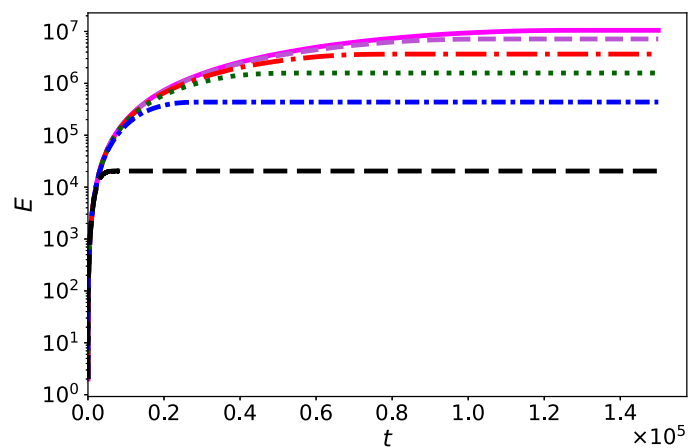


Fig. 8. Energy in the sandpile at each iteration for the rewired case, with 0 (magenta line), 50 (purple line), 100 (red line), 150 (green line), 200 (blue line), and 250 (black line) rewirings. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

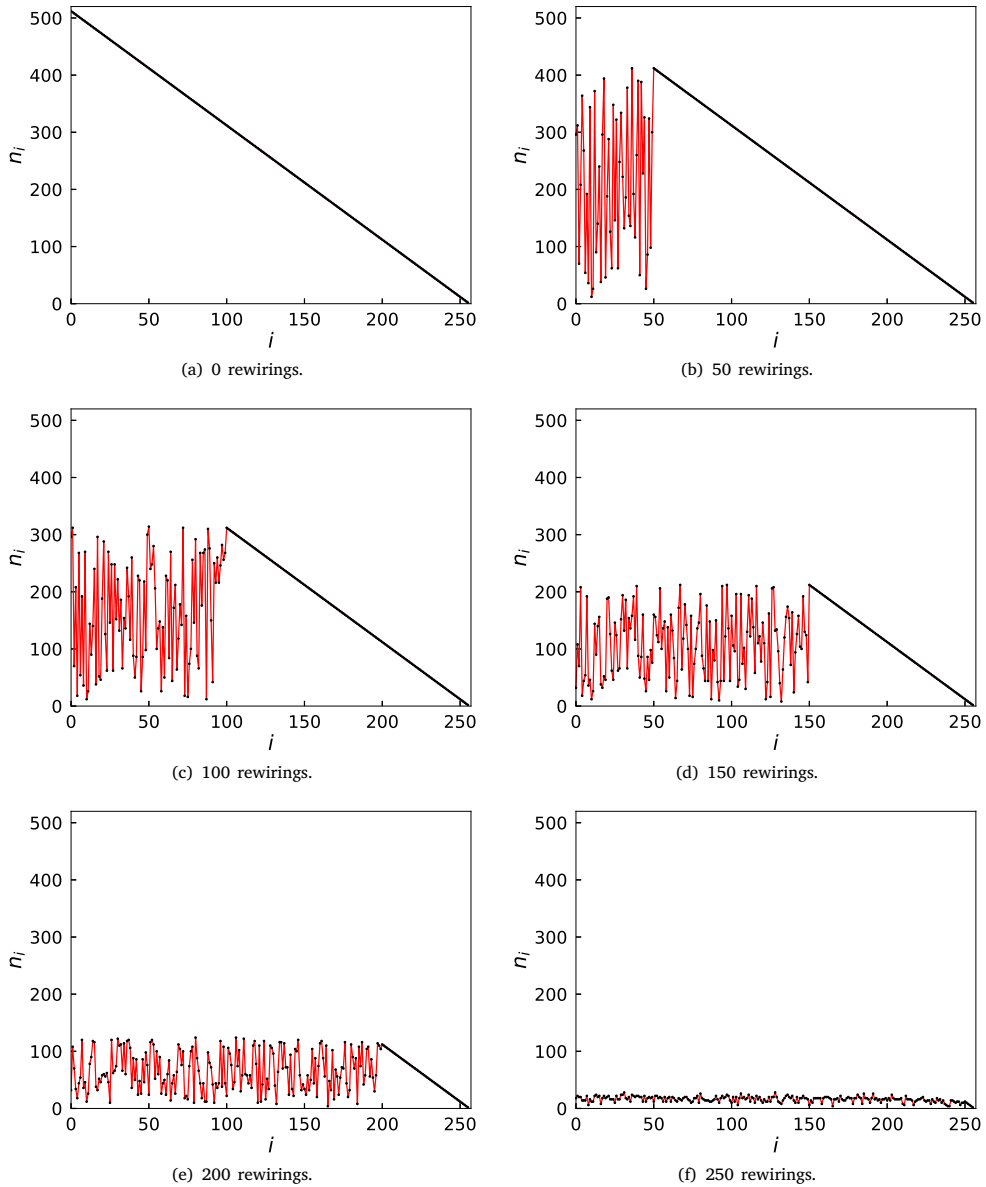


Fig. 9. Stationary states associated to the rewired networks, for $U = 2$, and $Q = 1$, and various numbers of rewirings: a 0 rewirings, (b) 50, (c) 100, (d) 150, (e) 200 and (f) 250.

As can be seen in Fig. 8, this change in topology causes the total energy of the rewired system to reach the critical state sooner; thus, when analyzing the avalanche statistics, the transient is always shorter than that of the original BTW system (This applies to both 1D and 2D).

The changes in the profile of the stationary states also modify avalanche statistics. This is shown in Fig. 10, where the PDF of the energy of avalanche events are plotted, for networks with the same number of rewirings as in Figs. 8 and 9. Since rewirings are selected randomly, we have taken 30 different sequences of rewirings, consistent with the rules in Section 2, and averaged their results. Thus, the curves in Fig. 10 are the average over 30 runs of different networks with 0 rewirings (magenta curve), 50 rewirings (purple curve), and so on.

Consistent with Figs. 8 and 9, it can be observed that large events are less likely as more rewirings are introduced. Also, avalanches occur in a narrower range of energies, consistent with the fact that it is no longer possible for a single site to accumulate a large energy anyway, as in the simple linear chain (Fig. 9(a)).

The complementary cumulative distribution functions $F(E)$ do not show a power-law scaling, consistent with the original BTW model, thus this particular feature is not modified by successive rewirings (see the Supplementary Material, Fig. S5). However, one

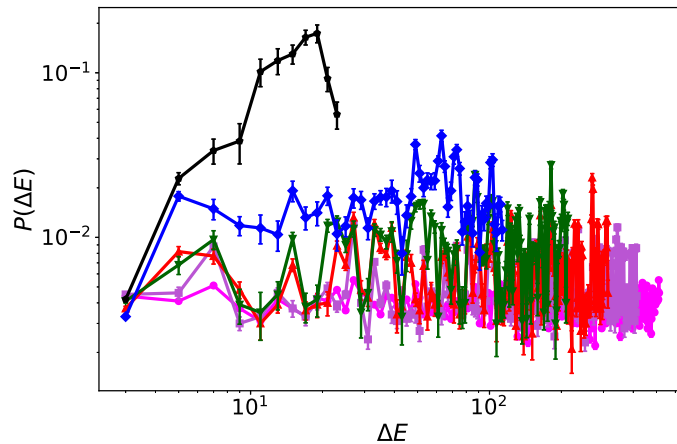


Fig. 10. PDF of energy release events for the rewired networks, averaged over 30 simulations with the same load $Q = 1$ and threshold $U = 2$, each simulation corresponding to a certain sequence of rewirings. Line styles represent different numbers of rewirings: 0 (magenta line, with circles), 50 (purple line, with squares), 100 (red line, with triangles), 150 (green line, with inverted triangles), 200 (blue line, with diamonds), and 250 (black line, with pentagons) rewirings. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

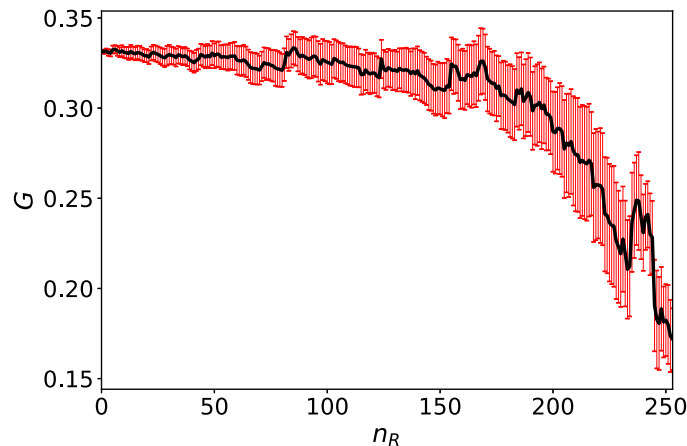


Fig. 11. Gini coefficient for the distribution of load in the stationary state, averaged over 30 samplings of the random sequence of rewiring. Error bars correspond to the standard deviation.

metric that shows a strong dependence on the number of rewirings is the Gini coefficient (G) of the load across the sandpile. This is shown in Fig. 11, where, again, 30 different random sequences of rewirings were averaged. It is clear that, up to 150 rewirings, the inequality of the distribution of load in the network is approximately constant, and then inequality rapidly decreases. The transition occurs at approximately 85% of the rewiring process. In Section 4 this is discussed in more detail, where size effects are considered.

Although the percolation argument makes sense, since more paths are available for the discharge, it is interesting to notice that this is not necessarily correlated to other measures of connectivity between nodes. A simple example is the average shortest path between nodes, defined as

$$\langle L \rangle = \sum_{s,t,s \neq t} \frac{d(s,t)}{n(n-1)},$$

where $d(s,t)$ is the shortest path from node s to node t , n is the number of nodes in the network, and $d(s,t) \equiv 0$ if there is no path from s to t . As seen in Fig. 12, the distance between nodes decreases as the number of rewirings increases, which is expected, but the decrease is much faster than for the Gini coefficient, and no particular transition is found.

We have carried out a similar analysis but changing the threshold to all integer values from $U = 2$ up to $U = 7$, in order to determine how results are affected by the specific value of the threshold chosen. These results, which consider different load values Q and threshold values U to study the robustness of the method, are included in the Supplementary Material. Figure S1 in the Supplementary Material shows the complementary cumulative distribution functions for various values of the threshold U , for load $Q = 1$, when the maximum number of rewirings has been reached. The main effect is the broadening of the distribution, as larger

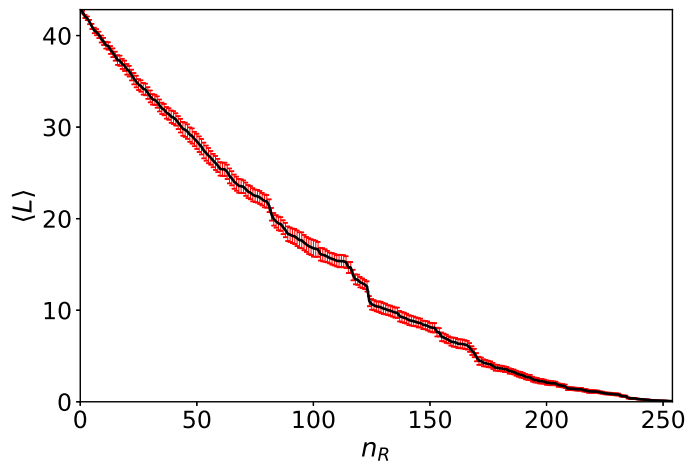


Fig. 12. Average distance between nodes as a function of the number of rewirings, averaged over 30 sequences of rewirings.

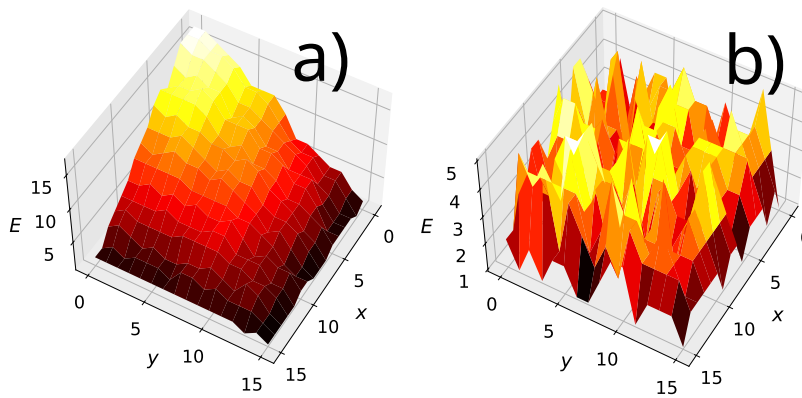


Fig. 13. Critical state for the $N = 16 \times 16$ system (a) without rewirings, (b) after 400 rewirings. The color map represents the number of grains per site.

values of the released energy in each avalanche are possible, which is expected if the load is constant, but a larger threshold is needed to trigger an avalanche. No clear exponential or polynomial tails are observed.

3.2. 2D case

For the 2D system, we performed simulations with 256 nodes (a 16×16 matrix), on which 30 different simulations were performed. Each simulation involves 480 rewirings and 10^5 iterations. The system starts from a steady state, using driver $Q = 1$ and threshold $U = 3$ (See Fig. 13).

In this case, after the stationary state is reached, the energy distributions for avalanches is a power-law, as seen in Fig. 14, which is consistent with Ref. [1], and also holds for the rewired network. The complementary CDF of the energy variations is shown in the Supplementary Material, Fig. S6.

A linear fit of the curves yields a power-law with exponents $\gamma \sim 1.45$, slightly increasing when the number of rewirings is large. Details are given in Fig. 15.

Unlike the 1D case, the Gini coefficient (see Fig. 16) does not present a sharp transition as the number of rewirings increases. Rather, it decreases until saturating at a minimum value after about 400 rewirings. This occurs because, for such large values of the number of rewirings, nodes that remain to be rewired are located at the edges of the system, and those nodes have only one connection which can be modified. This leads to the change in behavior observed in Fig. 16, as further modification of node inequality is slower.

As in the 1D case, the shortest average path does not show a transition between two behaviors (Fig. 17), but now it decreases at a linear rate with the number of rewirings n_R . The local maxima in Fig. 17 are due to the way the network is updated, following the labeling of the nodes, and because the change in average path length is different when rewiring involve nodes in the bulk or in the boundaries of the grid. In effect, since nodes at the boundaries must have one connection leading out of the system to ensure

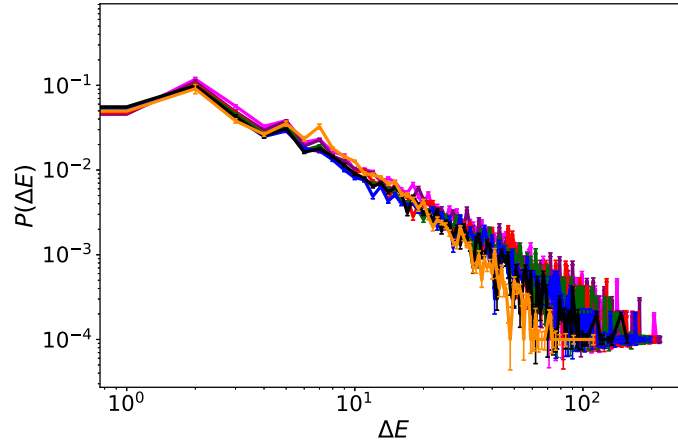


Fig. 14. PDF of released energy for different rewirings, in logarithmic scale, once the stationary critical state is reached. Colors correspond to 0 (magenta line), 80 (purple line), 160 (red line), 240 (green line), 320 (blue line), 400 (black line), 480 (brown line) rewirings. Charge is $Q = 1$ and threshold is $U = 3$. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

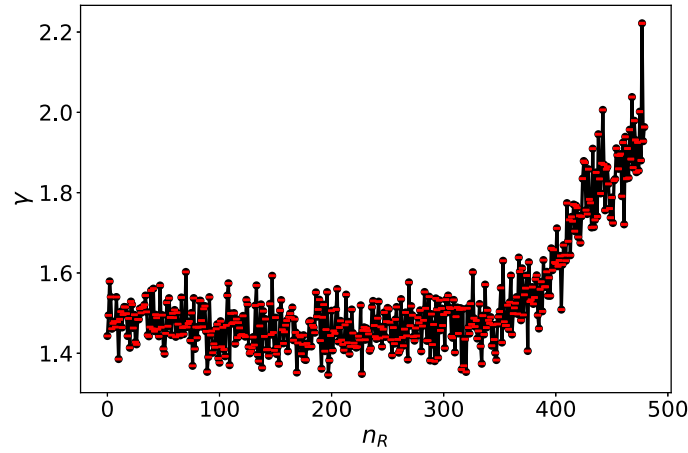


Fig. 15. Critical exponents γ for the PDF of dissipated energy in Fig. 14, as a function of the number of rewirings n_R . The system consists of a 16×16 grid, with n_R rewirings, and 10^5 iterations of the sandpile model after the critical state is reached. Then, results are averaged over 30 different sequences of n_R rewirings.

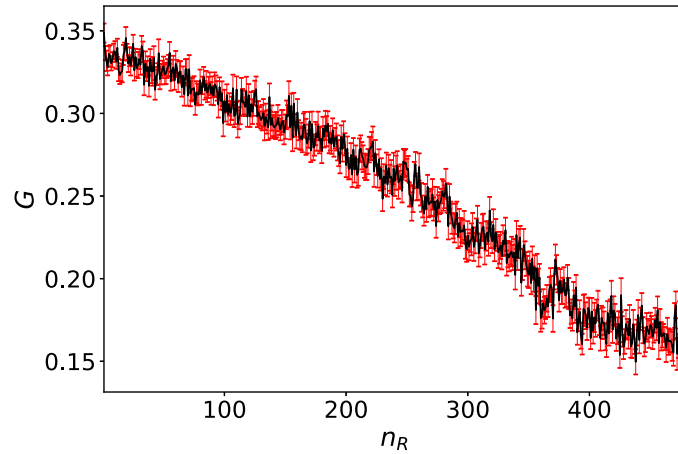


Fig. 16. Gini coefficient for the distribution of energy in the critical state, as a function of the number of rewirings. The system consists of a 16×16 grid, with n_R rewirings, and 10^5 iterations of the sandpile model after the critical state is reached. Then, results are averaged over 30 different sequences of n_R rewirings.

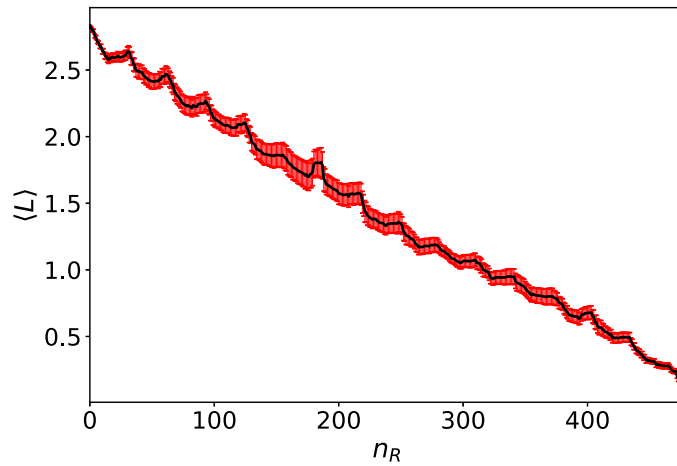


Fig. 17. Average shortest path length of the network as a function of the number of rewirings. The system consists of a 16×16 grid, with n_R rewirings. Then, results are averaged over 30 different sequences of n_R rewirings.

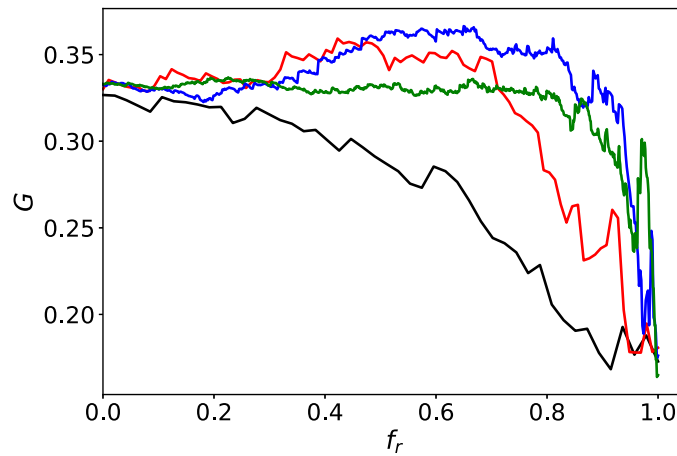


Fig. 18. Gini coefficient for the distribution of energy in the critical state of the 1D model, as a function of the fraction of rewirings f_r . Each curve corresponds to a different system size: $N = 50$ (black line), $N = 100$ (red line), $N = 500$ (blue line), and $N = 1000$ nodes (green line). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

energy dissipation, rewiring does not have the same effect for such nodes, as compared with nodes in the bulk. This is seen as local maxima of the average path length every time one row of the grid is completely rewired. Notice that, as mentioned in Section 2, our simulations have considered a square grid with 16 nodes per side, which is consistent with the 16 maxima observed in Fig. 17.

4. Size effects

The results in the previous sections show a clear transition in the Gini coefficient for 1D, but not in 2D, whereas there is a transition for the critical exponent for 2D, but not 1D. To better study this, and its possible dependence on size effects, we will study the evolution of the Gini coefficient as a function of

$$f_r = n_r / N_c,$$

where n_r is the number of rewirings, and N_c is the number of connections.

4.1. 1D case

In this case, we analyze the Gini coefficient for various network sizes, with $N = 50, 100, 500, 1000$ nodes, with $U = 2$ and $Q = 1$. We perform one simulation for each case, so curves are noisier than in the previous section, but a general trend can be appreciated. This is shown in Fig. 18.

Table 1

Critical value f_r^c at which the Gini coefficient has maximum derivative, for various number of nodes N .

N	f_r^c
50	0.6809
100	0.8247
500	0.8471
1000	0.8766

It is observed that, as the network size increases, the Gini coefficient tends to increase for all values of f_r , remaining approximately constant, until it collapses to a minimum value when all connections have been rewired. This implies that a sharper transition actually occurs for the Gini coefficient in the 1D case, but it is more evident for larger networks. The particular result previously shown in Fig. 11 is consistent with this.

We take the first derivative of G , (dG/df_r) , to determine where it has a maximum, which signals the critical value f_r^c at which the transition occurs. To perform a robust numerical calculation of the derivative, given the stochastic fluctuations of the curves, it is necessary to reduce the noise prior to differentiation. For this, we apply a Savitzky-Golay filter, which smooths the data by fitting a polynomial to a subset of adjacent data points. In our approach, a third-degree polynomial is used in conjunction with a sliding, overlapping window of dynamic width. To preserve the sharpness of the transition, the window size is set to 1/6 of the total data points for smaller datasets, up to a maximum fixed size of 21 points for datasets larger than 126 points (always constrained to an odd integer, as required by the algorithm). The results associated with the number of nodes and the control parameter at which the derivative is maximum are presented in Table 1.

We find that the transition point depends on system size, but for large N it stabilizes at around 85% rewiring, which is consistent with Fig. 11.

To analyze this transition from a topological perspective, we examine some network metrics. In Section 3 it was already found that the average shortest path length does not reveal a sharp signature of transition. Another metric related to shortest paths is the betweenness centrality $g(v)$, defined as

$$g(v) = \sum_{s,t \in V} \frac{\sigma(s,t|v)}{\sigma(s,t)},$$

where V is the set of nodes, $\sigma(s,t)$ is the number of shortest paths from node s to node t , and $\sigma(s,t|v)$ is the number of those paths passing through a given node v (other than s and t). This metric quantifies the global influence of a node over the energy flow.

Furthermore, we examine the in-degree ($\deg_{in}$) of the network. Because the total number of nodes and directed links is conserved during rewiring process, global averages such as the mean in-degree remain constant. Therefore, to capture how topological heterogeneity emerges, we focus on the variance and the maximum value of these structural metrics. This allows to quantify the inhomogeneity, and the possible concentration of incoming connections for some nodes.

The evolution of these structural features is presented in Fig. 19, which shows the maximum value and variance for both the in-degree (top panel) and betweenness centrality (bottom panel) as a function of the control parameter f_r .

From Fig. 19, it is observed that both the betweenness centrality and its variance tend to zero as the rewiring increases. These means that there are essentially no preferred shortest paths beyond the critical point for the transition of the Gini coefficient, unlike the extreme case of the initial linear chain configuration, where there is only one path for all energy releases.

As to the in-degree, figures show a slow increase of both the degree and the variance. Notice that degree can only have integer values, which explains the steps in the upper left panel. This behavior is consistent with the network becoming less homogeneous as rewiring increases.

4.2. 2D case

For two dimensions, Figs. 16 and 15 showed a transition for the critical exponent, but not for the Gini coefficient. To consider size effects, we take networks with $N = 16 \times 16$, $N = 24 \times 24$, $N = 32 \times 32$, and $N = 40 \times 40$ nodes, with $U = 3$ and $Q = 1$. As in the 1D case, one simulation per network was performed, but it is enough to see the general trend.

The Gini coefficient for the grain distribution in the network is shown in Fig. 20.

At first glance, the Gini coefficient does not show the abrupt break that the 1D system has, regardless of network size. However, there is a change of behavior at large f_r . Smoothing the curves as in the 1D case, and calculating the maximum absolute value of the derivative we find the corresponding critical value of the rewired proportion, f_r^c , at which this transition occurs. This is shown in Table 2.

We notice a similar behavior to that observed for the 1D case, in Table 1. In effect, Table 2 also shows that the critical point has a weak dependence on system size, approaching $\sim 85\%$ rewiring for large networks. The transition, however, is different, as it goes from a monotonously decreasing behavior to an essentially constant, or slightly increasing, rather than from a constant behavior to a rapidly decreasing one (Fig. 18). It would be interesting to further explore the evolution of the transition point, for larger network sizes. However, based on our numerical experiments, one should expect that it should increase at a smaller rate, since N is unbounded, whereas $f_r \leq 1$.

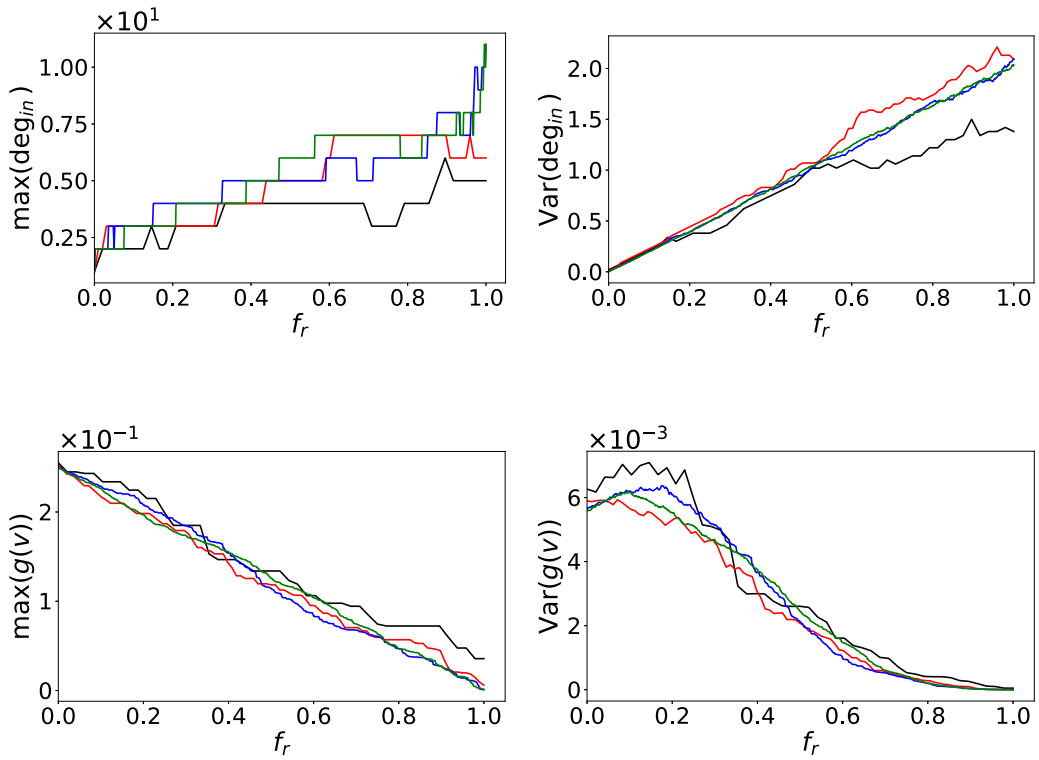


Fig. 19. Maximum values and variance of the in-degree and the betweenness centrality, as a function of percentage of rewiring, for various network sizes. Top panel: in-degree; bottom panel: betweenness centrality; left panel: maximum value; right panel: variance. Network size: $N = 50$ nodes (black line), $N = 100$ (red line), $N = 500$ (blue line), and $N = 1000$ (green line). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

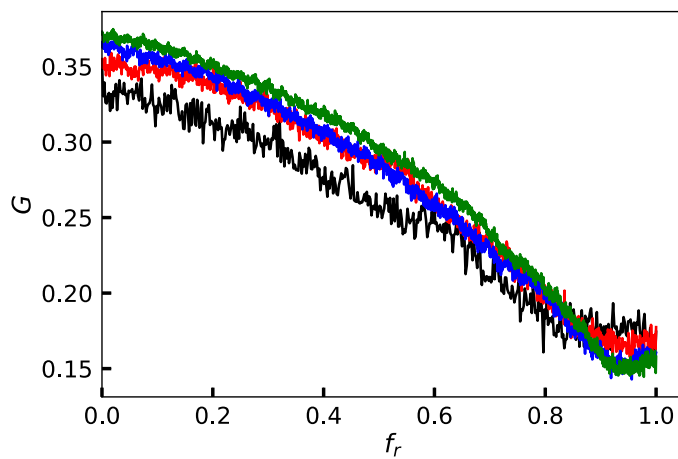


Fig. 20. Gini coefficient as a function of the percentage of rewired links f_r , for different system sizes N : $N = 16$ nodes (black line), $N = 24$ (red line), $N = 32$ (blue line), and $N = 40$ (green line). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 2

Critical value f_r^c at which the Gini coefficient has maximum derivative, for various number of nodes N , 2D case.

N	f_r^c
16×16	0.7107
24×24	0.7802
32×32	0.8390
40×40	0.8771

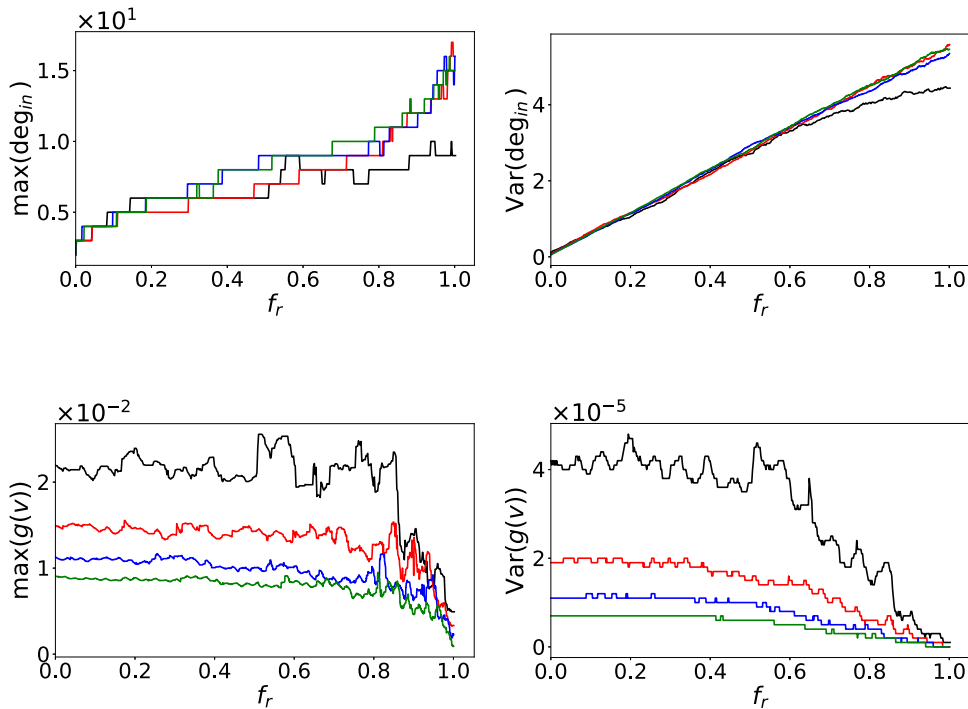


Fig. 21. Maximum values and variance of the in-degree and the betweenness centrality, as a function of percentage of rewiring, for various network sizes. Top panel: in-degree; bottom panel: betweenness centrality; left panel: maximum value; right panel: variance. Network size: $N = 16 \times 16$ (black line), $N = 24 \times 24$ (red line), $N = 32 \times 32$ (blue line), and $N = 40 \times 40$ (green line). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Fig. 21 is the analogous to Fig. 19, showing the maximum value and the variance of the in-degree and the betweenness centrality.

The in-degree behavior is similar to the 1D case. As to the betweenness centrality, both the maximum and the variance decrease, but after remaining essentially constant until, approximately, reaching the critical value for f_r . This result is consistent with that shown in Fig. 16, where the Gini coefficient decreases monotonically until it reaches a point where it flattens into a nearly constant curve.

5. Summary

In this work, we have studied, in a systematic way, the transition from the simple BTW sandpile to a sandpile over a directed network, where grains are added on the nodes, and grains are displaced to connected nodes following the directed edges. In general, choosing an arbitrary underlying network poses the risk of generating isolated nodes which accumulate loads instead of discharging them and closed topological loops. Both configurations trap energy, preventing the system from properly dissipating it even if the avalanche condition is locally satisfied. To address this issue, the model could be extended so that whenever a loop or an island emerges, one of the involved nodes is dynamically transformed into a sink. While this approach opens the door to more exotic network topologies, the present study focuses on a simple rewiring rule that inherently preserves the global degree and nodes and preserving SOC properties without the need to introduce new sinks. Thus, we randomly rewire nodes in such way that, at any stage of the network's evolution, energy release is always guaranteed once an avalanche is triggered. This constraint naturally leads to a tree-like structure with a single output node in 1D and with $2N - 2$ output nodes in 2D, and where all nodes have a single discharge path, while receiving discharges from several other nodes, depending on how the random rewirings occur. Then, we have

studied various features of the resulting stationary state, and of the avalanche statistics, to determine how they are modified as the “distance” between the network and the BTW model increases, quantified by the number of rewirings performed.

As more rewirings are introduced in the system, the number of discharge paths are increased, leading to shorter times to reach the stationary state. Also, since discharges can follow more than one path, the threshold condition to trigger the avalanches can be satisfied earlier than for the linear chain representing the BTW model, reducing the average load of nodes in the stationary state.

No power law or exponential behavior is found for the PDF of avalanche energy release events in 1D, whereas in 2D, a power law is found. These results are consistent with the original BTW model, and now we show that they persist even when the system is fully rewired.

An interesting behavior is observed when the load distribution across the 1D network is characterized by the Gini coefficient. Then, a clear transition is observed at about 85% the number of rewirings: for fewer rewirings, the Gini coefficient is essentially constant, and then, a sharp decrease is found. As we show that by studying the behavior of the system as a function of the proportion of rewired nodes f_r , and changing the network size. Here, $f_r = 0$ represents the initial, nonrewired system, and $f_r = 1$ is the case where all connections have been modified. We find that dG/df_r has a maximum around 85% rewiring. This is due to the easier distribution of load between nodes due to the larger number of discharge paths, but the transition is not evident by observing other features of the avalanche events, such as the PDF of energy releases, or metrics of the underlying network topology, such as the average distance between nodes, which decreases monotonically as the number of rewirings increases, without any particular transition. However, betweenness centrality, which is also related to shortest paths, but in a less trivial way, does indeed show a transition, where it is essentially null for all nodes, suggesting that there are no preferred nodes to connect two random nodes of the network. This occurs, as seen in Fig. 19, after 85% rewiring, consistent with the transition observed for the load distribution.

In the 2D case, a transition in the Gini coefficient occurs at about 400 rewirings in our case. It is interesting that this correlates with the behavior for the power-law exponent for the energy distributions, which remains at $\gamma \sim 1.45$, except after 400 rewirings, where it increases up to 2.21. This observation is further examined by the analysis of size effects, where it is shown that a smooth transition does indeed occur in the Gini coefficient, for different network sizes. Thus, the value of 400 found for the particular case studied in Section 3 is consistent with the critical value $f_r^c \approx 0.85$, as can be seen in Section 4.

We also notice that dimensionality has various consequences that can explain the fact that the transition is smoother for 2D with respect to the 1D case. As already observed in the 1D case, rewiring adds new discharge paths that decrease the energy of the stationary state, and the avalanche size. However, since the boundary is not changed, transport flows to the same final node. In the 2D case, rewiring has the same effect on stored and released energy, but more exit paths are available, since energy can be released at any boundary node. Thus, on average, modifying a particular connection should not have a strong effect on the distribution of load, and thus on the Gini coefficient due to the larger availability of discharge paths in the 2D system.

Our work suggests that the Gini coefficient of the grain distribution can be a useful way to examine the transition of transport properties in a complex network. This metric, typically associated to the measurement of inequality in economics, is a useful way to characterize the distribution of a variable across a system by means of a single scalar value. In general, any scalar metric derived from the distribution function has less information, and thus the issue is whether the metric provides any useful insight. In our case, previous works had already shown the relevance of the Gini coefficient to yield information about the evolution of a network of sunspots along the solar cycle, unlike other usual metrics such as the network density or the clustering coefficient [26], and to reveal the transitions between the elastic and plastic regimes during deformations in metallic glass [37], transitions which are not clear in the distribution functions themselves.

In this sense, it is interesting to observe that transitions in the Gini coefficient of relevant variables have also been observed in other types of networks, such as in power grid networks where connections are removed as part of a robustness analysis of the network [38], where the Gini coefficient shifts from a range of $G = 0.8$ to $G = 0.6$ to $G \approx 0.32$. Other studies have considered systems more similar to ours, such as percolation on a square lattice, and the fiber bundle model of fractures [39], also finding that the Gini coefficient reveals phase transitions in such systems.

As to the shortest path, it decreases as in the 1D case, no transition is observed either, but the decrease is linear with the number of rewirings, except for sudden and local increases at specific values of n_R , which are related to the nodes where released from the system at the edges of the grid. These nodes have the fewest connections within the system since only one connection is directly connected to the network, while the rest of the nodes have two connections.

Rewirings in this model affect the topology of the network and, in turn, its transport properties. These results could be relevant to study energy release along preferential paths, such as in magnetic reconnection events, thunderstorms, or money flow, processes where a certain variable is transferred between agents or locations which are topologically, but not necessarily spatially close to each other.

Finally, it is interesting to note that the exponents obtained for energy release are similar with some obtained by models and observations of solar flares, which range between $\gamma = 1.44$ and $\gamma = 2$ depending on the type of model and measurement technique [4,6,23]; and in electrical discharges where the release of energy follows a power law with exponent $\gamma = 2.3$ [40]. These exponents for the two dimensional case, in addition to being consistent and falling within the range of sandpile models and these examples of physical systems, exhibit a saturation of approximately 85%, just like the two-dimensional Gini coefficient. This effect, both in the case of the Gini coefficient and in the case of the avalanche energy dissipation exponents, is explained by the fact that nodes closer to the last sink node are more likely to acquire new connections. Consequently, highly connected topological hubs emerge of these hubs while other become depleted. As f_r increases, the in-degree of these hubs grows considerably, as shown in Figs. 19 and 21.

This localized accumulation of connections has two fundamental dynamical effects. First, these hub nodes gain multiple energy input pathways, causing them to reach their critical threshold and subsequently release energy much more rapidly. Second, this rewiring alters the global energy transport, as evidenced by the betweenness centrality. As the rewiring fraction approaches the critical regime ($f_r \sim f_r^c$), the variance of the betweenness centrality rapidly drops towards to zero. This vanishing variance alongside a decreasing maximum betweenness signals a topological homogenization where, on average, all nodes contribute equally to energy flow, leading to a more direct and efficient energy flow through shorter escape paths to the system boundaries.

The present analysis could be useful for systems which both exhibit SOC features, and whose transport flow is directed on a nonhomogeneous network, as is the case of magnetic field lines directing flow from one magnetic active region to the opposite region, river basins from the mountains to the sea, or electrical discharges between clouds and the ground. Different network models and rewiring strategies should be necessary to study such systems, but previous work based on the Lu-Hamilton model for solar flares [23] suggests that rewired networks pose interesting issues for the study of complex systems exhibiting SOC features.

CRediT authorship contribution statement

Alejandro Zamorano: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Víctor Muñoz:** Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.chaos.2026.118629>.

Data availability

No data was used for the research described in the article.

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